

AI-Based Paddy Seed Quality Assessment And Farmer Feedback System

*A thesis submitted in partial fulfilment of the requirements
for the award of the degree of*

Bachelor of Technology

By

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Assistant Professor



**Department of
Computer Science & Engineering**

Manipur Technical University

May, 2026

DEDICATION

This thesis is dedicated to all those who have inspired, supported and stood by us throughout the journey of this project.

To our families: Your unwavering love, patience, and encouragement have been the bedrock of our emotional strength during the most challenging phases. *Your belief in our potential has fortified our resilience and determination.*

To our mentors and instructors: Your expertise, thoughtful guidance, and commitment to our academic growth have been instrumental in shaping both this project and our understanding of the field. *Your encouragement and constructive feedback have been invaluable at each stage.*

To our friends and peers: Your camaraderie, moral support, and motivation have illuminated our path. *Your presence transformed moments of stress into opportunities for shared growth and learning.*

To the global community of researchers, engineers, and innovators in Artificial Intelligence and Computer Vision: Your pioneering work has laid the foundation for this thesis. *Your dedication to advancing knowledge continues to inspire and propel us to contribute meaningfully to this dynamic field.*

This thesis stands as a testament to the collective support, shared vision, and unwavering encouragement of all who have been part of our academic and personal journey.

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BONAFIDE CERTIFICATE

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to the **Department of Computer Science & Engineering**, in partial fulfilment of the requirements for the award of the **Bachelor of Technology** degree in **Computer Science and Engineering**, is a genuine and original work carried out by them under my supervision during the academic year **2025–2026**.

The project embodies the results of their own research and development work and has not been submitted, either in part or in full, for the award of any other degree or diploma of this university or any other institution. The work has been carried out with sincerity, academic integrity, and adherence to research ethics.

We consider this work worthy of consideration for the partial fulfillment of the requirements for the said degree.

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DECLARATION

I hereby declare that the thesis entitled “**AI-Based Paddy Seed Quality Assessment And Farmer Feedback System**”, submitted to the **Department of Computer Science and Engineering, Manipur Technical University**, in partial fulfilment of the requirements for the award of the **Bachelor of Technology** degree, is a result of my own work and effort. Under the guidance of **Dr. Roshni Rajkumari**, Assistant Professor, Department of Computer Science and Engineering, Manipur Technical University.

I affirm that this work has been carried out with academic honesty and integrity. All sources of information have been properly cited and acknowledged. This thesis has not been submitted earlier, either in part or in full, for the award of any degree or diploma at this or any other institution.

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Our heartfelt thanks go to our project supervisor, **Dr. Roshni Rajkumari**, Assistant Professor, whose consistent guidance, insightful feedback and unwavering support played a pivotal role in shaping this project. We also extend our sincere appreciation to **Mrs. Tayenjam Aerena**, Head of Department, for her inspiring leadership and motivation.

We are also thankful to all the faculty members, technical staff and friends who provided their valuable input and moral support throughout the journey.

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ABSTRACT

The quality of paddy seed plays a vital role in crop yield, germination rate and overall agricultural productivity. Many small and marginal farmers suffer crop loss due to substandard or counterfeit seeds that closely resembles certified products. Due to lack of affordable and rapid verification methods, farmers rely on packaging labels and basic manual inspection, which is time consuming, inconsistent and error prone. Poor quality seeds are generally known after the seeds are sown, and thus, results in low germination, reduce yield and incur a financial loss.

To address these challenges, our project entitled "**AI-Based Paddy Seed Quality Assessment And Farmer Feedback System**" is designed to combat the spread of counterfeit and substandard paddy seeds. This project is a web-based platform that utilizes artificial intelligence and deep learning model to automate the classification of quality seeds into healthy and suspicious ones based on characteristics like size, shape uniformity, colour, surface texture and visible defects. Farmers can scan seed samples using a simple image-based interface to receive an estimated quality assessment of paddy seeds, presented in clear, farmer friendly language.

In addition to seed quality assessment, our system also includes a Farmer Complaint and Feedback module that enables reporting of field level issues such as poor germination or crop failure, cultivation experience, seed performance, etc.

The system avoids chemical analysis, specialized hardware or complex supply chain systems, making it low-cost, scalable and suitable for rural deployment. It also helps farmers select better quality seeds, reduce crop failure, and improve productivity. The integration of artificial intelligence with farmer feedback system makes the proposed system a useful tool for modern agricultural practices and sustainable farming development.

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Chapter 1

Introduction

1.1 Background and Context

Agriculture is the foundational backbone of many global economies, heavily influencing food security, employment and national income. At the core of all agricultural productivity depends on the quality of the seed planted, which is the most critical and non-negotiable factor determining crop yield, resilience to disease and overall harvest quality. However, the agricultural sector is currently facing a severe crisis due to the increasing problem of counterfeit or fake seeds in the market.

Counterfeit seeds are often packaged to resemble premium brands or are visually indistinguishable from high-yield varieties, yet they exhibit drastically lower germination rates and poor genetic traits. Traditional methods of verifying seed authenticity rely on destructive laboratory testing or subjective physical inspection, both of which are inaccessible, expensive and far too slow for the an average farmer. Consequently, farmers often discover that they have been deceived only after planting, leading to catastrophic losses in time, resources and income.

1.2 Motivation

The primary motivation for this project stems from the devastating socio-economic impact that low-quality seeds have on farming communities. When farmers unknowingly purchase and plant low-quality seeds, they face severe financial distress, which cascades into broader issues of local food insecurity and loss of trust in agricultural supply chains. There is an urgent, unmet need for accessible technological tools that empower grassroots agricultural workers to protect

themselves against fraud.

Simultaneously, the rapid advancement of deep learning, computer vision, and cloud computing presents an unprecedented opportunity to address this crisis. By leveraging deep learning and computer vision techniques it is now possible to achieve automated, laboratory-grade visual analysis of seeds. The motivation is to harness these cutting-edge AI technologies and package them into a simple, web-based tool, putting the power of automated, real-time seed verification directly into the hands of farmers.

1.3 Problem Statement

Despite the devastating effects of agricultural fraud, farmers currently lack a reliable, accessible, and rapid method to accurately distinguish between genuine and counterfeit seeds before the planting season begins. Visual similarities make manual detection nearly impossible and existing institutional supply chains lack a transparent, real-time verification system.

The core problem is the absence of an integrated, technology-driven solution that can automatically analyze seed imagery, identify fraudulent anomalies and capture feedback from farmers to map and identify the spread of counterfeit seeds in real-time.

1.4 Objectives

The main goal of this project is to create an intelligent ecosystem for seed verification. To achieve this, the project outlines the following specific objectives:

1. To design and train a deep learning model capable of accurately classifying seed images and identifying anomalies associated with counterfeit seeds.
2. To develop and deploy a user-friendly web-based platform that allows farmers to upload images of their seeds for AI inference and verification.
3. To integrate a "Farmer Feedback System" within the platform, enabling users to report their post-purchase and post-planting experiences (such as germination failure rates).
4. To utilize the feedback and newly uploaded images to continuously refine the database, creating an early warning system against emerging counterfeit supply chains.

1.5 Scope of the Project

The scope of this project encompasses the end-to-end development of an AI-driven web application tailored for agricultural quality control. This includes the collection, curation, and preprocessing of a targeted seed image dataset, followed by the training, validation, and optimization of a Convolutional Neural Network (CNN).

Additionally, the scope covers the full-stack development of the web platform, including an intuitive frontend interface for image uploads and a robust backend to handle model inference and database management. The project is strictly limited to the non-destructive, visual assessment of seeds via digital imagery; it does not extend to biochemical, genetic, or DNA-based testing methods. The final deliverable will be a functional prototype that demonstrates the feasibility of combining deep learning with crowdsourced feedback to combat agricultural fraud.

Chapter 2

Literature Review

2.1 Seed Quality Assessment: Traditional Methods

Conventional seed quality assessment rests on two broad approaches: manual visual inspection and laboratory-based biochemical testing. In field practice, inspectors evaluate seeds by examining size, shape, colour and surface texture; judgements that are inherently subjective, vary between observers and resist standardisation across districts or seasons. Laboratory tests such as the Tetrazolium (TZ) test and standard germination assays offer more reliable viability estimates, but both are destructive and typically require four to seven days under controlled conditions. That turnaround makes them impractical for screening large consignments at intake points, or for farmers who need a decision before the planting window closes.

Near-infrared (NIR) spectroscopy and X-ray transmission imaging have been investigated as non-destructive alternatives. NIR detects internal moisture gradients and biochemical composition without damaging the sample; X-ray imaging can reveal embryo defects invisible on the seed surface. Both perform well in laboratory conditions, but equipment costs typically USD 15,000-80,000 per unit and the requirement for trained operators confine their use to national seed-testing stations. Smallholder farmers and district-level inspectors in regions such as northeast India have no practical access to either technology, which is precisely where the need for affordable, rapid screening is most acute.

2.2 Deep Learning for Agricultural Image Classification

Convolutional Neural Networks (CNNs) have been applied to a range of agricultural classification problems over the past decade, including leaf disease iden-

tification [1], weed recognition [2], and post-harvest quality grading [3]. Their principal advantage over classical machine-learning pipelines is that spatial features are learned directly from raw pixel data during training, removing the need to manually engineer feature extractors for each new crop or defect type.

Applying CNNs to seed classification follows naturally from this body of work. Paddy seeds exhibit subtle differences in surface texture and colour between certified and counterfeit lots; differences that resist quantification through handcrafted descriptors but emerge reliably from sufficiently large training sets. The computational cost of running a trained CNN at inference time is also modest enough to support server-side web deployment, which matters when target users have intermittent electricity but reasonable mobile-data coverage.

2.3 Transfer Learning and ResNet Architectures

Training a CNN from random initialisation requires on the order of tens of thousands of labelled examples per class to achieve stable convergence. Annotated seed image datasets of that scale do not yet exist for most crop varieties, and certainly not for the specific task of detecting counterfeits. Transfer learning is the standard response: a model pre-trained on ImageNet (1.28 million labelled images across 1,000 categories) is fine-tuned on the smaller domain-specific dataset. The early convolutional layers, which encode low-level features such as edges and textures, are retained; only the later task-specific layers are updated. In practice, this approach consistently reaches accuracy close to what a from-scratch model would achieve on a dataset roughly ten times larger.

ResNet architectures are well-suited to this fine-tuning scenario. The residual skip connections introduced by He et al. [4] allow gradients to propagate through very deep networks without vanishing, making it feasible to fine-tune models with 50 or more layers on datasets that would cause shallower architectures to overfit. ResNet-50 has become a common baseline in agricultural image classification, balancing parameter count (≈ 25 million) against accuracy on standard benchmarks, with pre-trained weights publicly available. For this project, it provides a well-validated starting point that reduces both training time and the risk of overfitting on the available paddy seed dataset.

2.4 Farmer Feedback Systems

A detection model trained on a fixed dataset will degrade over time if counterfeiting practices evolve and no new training data are collected. Field-level feedback from farmers is one practical mechanism for capturing these shifts. When a farmer submits a seed batch for image-based verification and subsequently reports germination failure or abnormal yield, that report constitutes a labelled example, a model prediction paired with a ground-truth outcome, that can be incorporated into periodic retraining cycles.

Feedback also serves a more immediate diagnostic function. Aggregating reports across users can flag a suspect lot or supplier before the model itself is updated: three independent failure reports from the same district within a fortnight is a meaningful signal, regardless of what the image classifier predicts. This rule-based alerting complements the model rather than replacing it. Sustaining farmer participation remains a practical difficulty; deployments in low-resource contexts consistently find that single-tap rating interfaces produce far higher response rates than free-text fields or structured forms [5].

2.5 Web-Based Agricultural Platforms

Deploying a trained model as a web service separates the computational workload from the end user’s device. A farmer uploads a seed image through a browser; the server runs inference and returns a result. This architecture offers two concrete advantages over distributing a standalone mobile application: the model can be updated centrally without requiring client-side installations, and every prediction is logged in a structured format that supports monitoring and retraining.

For farmers in Manipur and comparable regions, browser-based access via a mid-range Android handset is more realistic than assuming a dedicated application installation. Keeping the frontend lightweight, avoiding heavy JavaScript bundles and large media assets reduces page-load times on 4G connections with variable signal quality. Containerised inference endpoints handle computationally intensive tasks without requiring a permanently running server, reducing operating costs during off-season periods when query volume is low.

2.6 Review of Related Work

Early automated seed classification employed Support Vector Machines (SVMs) and Random Forests operating on handcrafted features derived from shape measurements, colour histograms, and texture descriptors. Ni et al. [6] classified rice seed varieties using morphological parameters extracted from binary silhouettes, achieving 94% accuracy across five varieties under controlled lighting. These methods perform adequately when intra-class variation is low and imaging conditions are uniform, but generalise poorly to photographs taken in field settings.

Deep learning approaches have largely displaced handcrafted methods in more recent work. Liu et al. [7] applied VGG16 transfer learning to maize seed defect detection, reporting 97.3% accuracy on a 12-class dataset. Xu et al. [8] used ResNet-50 for soybean quality grading and found that fine-tuning from ImageNet weights outperformed training from scratch by approximately 8 percentage points when training data were limited to fewer than 2,000 images per class. Both studies, however, address naturally occurring defects, such as cracking, mould and insect damage, rather than intentional substitution. A counterfeit seed is engineered to resemble the genuine article; distinguishing the two is a harder classification problem than separating a damaged kernel from an intact one. The visual gap between classes is narrower, and the adversarial nature of counterfeiting means the distribution of counterfeits may shift as producers adapt. No published study directly evaluates CNN performance on verified counterfeit-versus-genuine seed pairs, and none integrates a structured feedback collection mechanism into the evaluation pipeline.

2.7 Summary and Research Gaps

The reviewed literature establishes that transfer-learned CNNs are effective for agricultural image classification, including seed quality assessment. Three gaps are directly relevant to this work. First, no published study has targeted intentional counterfeiting as distinct from natural quality defects; the two problems differ in that counterfeits may pass casual visual inspection and require the model to detect subtle discriminative cues. Second, existing models are evaluated on laboratory-quality images collected under controlled conditions; performance on photographs taken by farmers using mobile devices in variable lighting has not been reported. Third, no end-to-end system has been published that connects model inference to a structured feedback collection mechanism and uses that feedback iteratively for retraining.

This project addresses all three gaps. The classifier is trained and evaluated specifically on a fake-versus-genuine paddy seed dataset; the web platform is tested with mobile-captured images; and the farmer feedback module is built directly into the application, with logged responses stored in a format suitable for future retraining rounds.

Chapter 3

System Design and Architecture

3.1 Overall System Architecture and Technology Stack

The AgriVerify platform is designed using a multi-tiered architecture based on the Clean Architecture model to ensure strict separation of concerns, decoupling of business logic from external frameworks and high maintainability [9]. The system comprises three primary layers: a Next.js frontend client web application, an ASP.NET Core backend Web API and a FastAPI deep learning inference microservice. These tiers interact to process seed validation requests, manage user authentication and persist data.

The backend API follows Onion Architecture principles, dividing responsibilities into four distinct concentric layers as detailed in Table 3.1.

Table 3.1: Backend Architectural Layers and Responsibilities

Architectural Layer	Core Responsibilities and Encapsulated Components
Domain Layer	Core domain entities (User, VerificationHistory), value objects, and domain exceptions. Completely independent of external frameworks.
Application Layer	Orchestration logic (VerificationService, UserService), Data Transfer Objects (DTOs), mapping configurations and request validation rules.
Infrastructure Layer	External integration implementations, including database persistence via Entity Framework Core, Azure Cognitive Services and Blob storage managers.
API Layer	REST API controllers, request-response handling, JWT authentication middleware and unified exception filters.

Table 3.2 details the software development kits (SDKs), frameworks, libraries, and cloud platforms employed across the frontend, backend, and machine learning components.

Table 3.2: Consolidated Technology Stack of AgriVerify

Component	Framework / Library	Functionality / Purpose
Frontend	Next.js 14 (TS)	Client interface, server-side rendering, and API proxy routing [10].
	TanStack Query	Server state synchronization and asynchronous data caching [11].
	Zustand	Client-side authentication state management [12].
	Tailwind CSS	Styling system and component customization [13].
Backend	ASP.NET Core 8	Orchestration Web API, routing, and controller layer [14].
	Entity Framework Core	Object-Relational Mapping (ORM) for SQL Server database access.
	Azure Cognitive Services	Optical Character Recognition (OCR) and Custom Vision packaging verification.
	Azure OpenAI Service	Natural language synthesis for seed verification reports.
ML	PyTorch 2.1	Deep learning framework for neural network training [15].
	ResNet-50	Pre-trained CNN backbone for seed variety classification [4].
	OpenCV	Preprocessing and morphometric DUS parameter analysis [16].
	FastAPI	Serverless microservice API on GCP Cloud Run [17].

The PyTorch ResNet-50 model is containerized and deployed as a FastAPI microservice on Google Cloud Platform (GCP) Cloud Run to enable automatic, serverless scaling [17]. During inference, input images undergo size standardization to 224×224 pixels and normalization using ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). OpenCV extracts 17 Distinctness, Uniformity, and

Stability (DUS) morphometric parameters, which are compared against reference variety profiles to compute similarity scores [18].

3.2 Functional and Non-Functional Requirements

System requirements are categorized into functional specifications, defining the core operations and roles within the application, and non-functional specifications, outlining quality attributes and system constraints. These requirements ensure that the platform meets both agricultural user needs and architectural standards.

The functional requirements (Table 3.3) define the core capabilities across different system actors (farmers, agricultural officers, and administrators). Concurrently, the non-functional requirements (Table 3.4) dictate operational performance, security, and portability parameters, ensuring standard enterprise scalability and reliability.

Table 3.3: System Functional Requirements

ID	Module / Role	Functional Specification
FR-01	Authentication	User registration and authentication via JWT for farmers, officers, and administrators.
FR-02	Image Submission	Multi-modal submission (packaging and seed image) for real-time verification.
FR-03	AI Inference	Execution of Custom Vision, OCR text extraction, and ResNet-50 DUS profiling.
FR-04	Feedback System	Reporting mechanism allowing farmers to submit grievances on counterfeit seed batches.
FR-05	Case Management	Dashboards for officers to track, update, and resolve seed complaints.
FR-06	Risk Analytics	Automated aggregation of regional complaints in a Batch Risk Registry for analytics.

Table 3.4: System Non-Functional Requirements

ID	Quality Attribute	Specification
NFR-01	Performance	Target end-to-end inference latency under 3.0 seconds through parallel service calls.
NFR-02	Scalability	Stateless API design facilitating horizontal scaling and microservice containment.
NFR-03	Security	HTTPS communication, JWT session control, role-based authorization (RBAC), and hashed secrets [19].
NFR-04	Reliability	Atomic database operations and soft-delete implementation for data auditability.
NFR-05	Usability	Responsive UI designed for mobile and desktop screens, optimized for low-bandwidth environments.

3.3 High-Level System Architecture

The system architecture is structured as a three-tier model: the Client Layer, the Backend Orchestration Tier, and the Cloud AI and Data Infrastructure Tier. As illustrated in Figure 3.1, client applications do not interact directly with the backend API; rather, all requests are routed through a server-side proxy in the Next.js frontend web tier. This proxy injects JSON Web Tokens (JWT) for authentication and forwards requests to the ASP.NET Core 8 Web API. The API layer orchestrates calls to the persistent data store (SQL Server), cloud storage, and AI inference services.

3.4 System Data Flow Pipeline

The step-by-step verification pipeline and data transmission flow are depicted in Figure 3.2. The execution sequence is structured as follows:

1. **Submission:** The client uploads multi-part image data representing seed packaging and physical seeds.

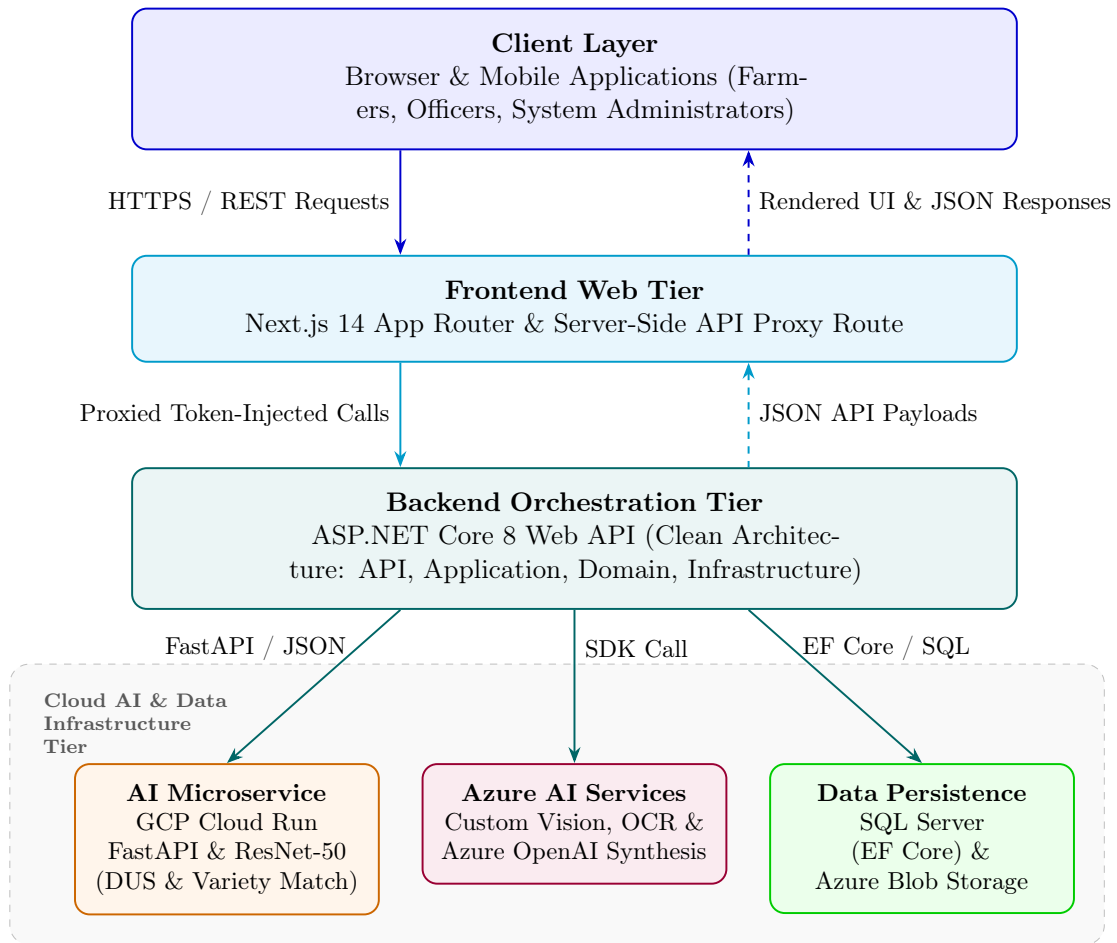


Figure 3.1: High-level system architecture of the AgriVerify platform.

2. **Routing and Proxying:** The Next.js API proxy interceptor validates user sessions, appends a secure bearer token, and executes an HTTP POST request to the backend.
3. **Parallel Execution:** The ASP.NET Core orchestration engine coordinates concurrent asynchronous requests using asynchronous task-based execution.
4. **Feature Extraction:**
 - Custom Vision classifies the packaging (Genuine, Counterfeit, Suspicious).
 - Azure AI Vision extracts text strings via OCR to identify seed batch codes.
 - The GCP Cloud Run ResNet-50 service determines 17 morphological parameters.
5. **Synthesis and Persistence:** The OpenAI integration consolidates these individual model predictions into a coherent diagnostic summary, which is stored in the relational database via Entity Framework Core before the final JSON payload is returned to the user interface.

3.5 Database Entity-Relationship Model

The relational database schema is modeled to ensure data consistency, minimize redundancy, and support real-time querying. Figure 3.3 defines the entity relations, key fields, and cardinalities within the AgriVerify database. The core architectural design decisions are detailed as follows:

- **User and Security Tracking:** The User entity supports role-based access control and maintains a 1-to-many relationship with RefreshToken to secure active sessions.
- **Verification History:** The VerificationHistory table preserves all client evaluations and maintains a soft-delete status (IsDeleted) to enforce an immutable audit trail.
- **Feedback and Case Resolution:** When a seed dispute is identified, a ProductComplaint record is initiated. This entity links to the executing officer, records ongoing investigations via ComplaintAction, and syncs batch metrics with the BatchRiskRegistry.

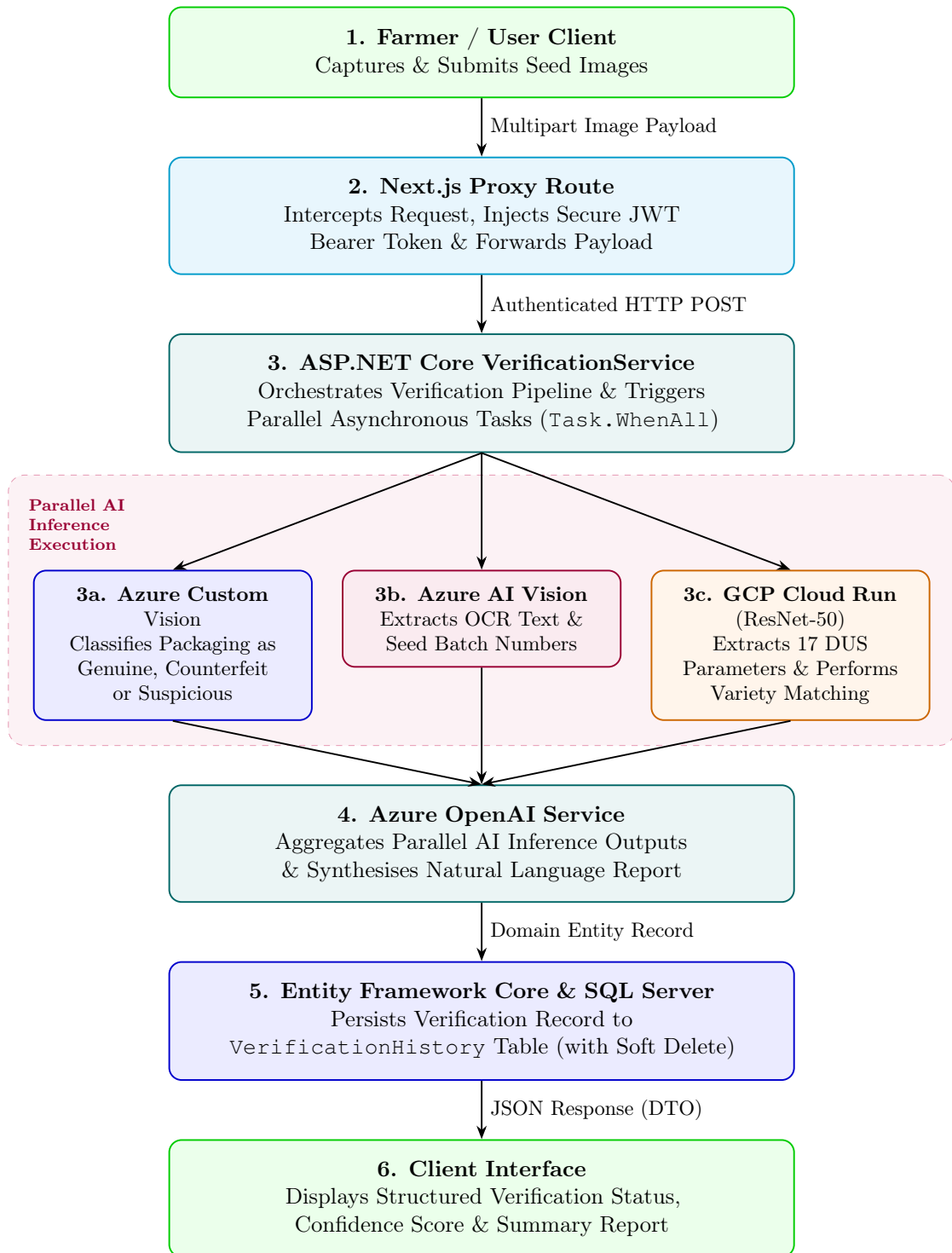


Figure 3.2: Data flow diagram for the seed verification pipeline

- **Batch Analytics:** The BatchRiskRegistry dynamically compiles aggregation data (complaint counts, severity indices, and regional distributions) to optimize query performance for system dashboards.

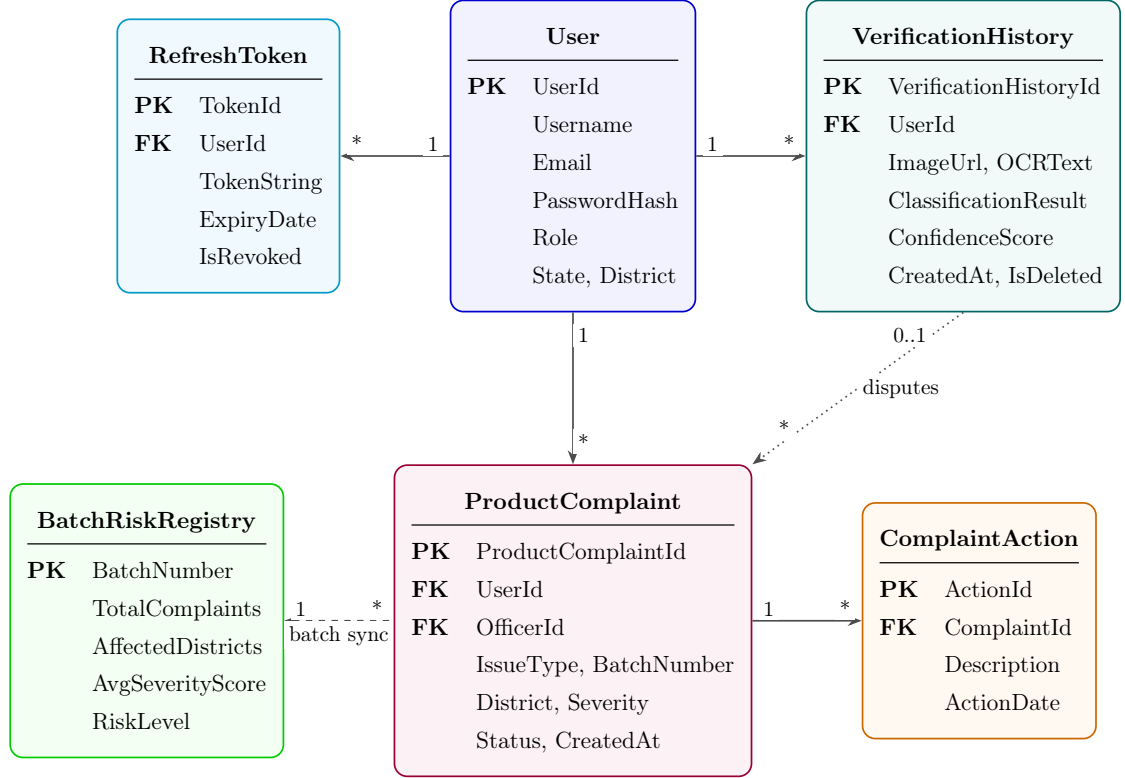


Figure 3.3: Simplified entity-relationship diagram for the AgriVerify domain model. Cardinality notation follows UML conventions.

Chapter 4

Model Serving API

Deploying a trained model as a live service introduces a different set of problems than training it. Model development focuses on architecture and accuracy; the serving layer determines response latency, concurrent throughput, and uptime under real load [20]. In AgriVerify, the PyTorch ResNet-50 classifier from Chapter 4 runs inside a dedicated inference microservice built specifically for this purpose.

This chapter describes the design of that microservice, implemented with FastAPI and Uvicorn [21]. It covers the asynchronous request lifecycle, the tensor normalization pipeline, the API endpoint contracts, and the containerised deployment approach using Docker multi-stage builds on GCP Cloud Run [22].

4.1 API Architecture and Request Orchestration

4.1.1 FastAPI and ASGI Serving Stack

The inference microservice runs on FastAPI atop the Uvicorn ASGI runtime. FastAPI was chosen for its native async I/O support, Pydantic-based request validation and automatic OpenAPI schema generation [21]. Uvicorn’s `uvloop`-backed event loop handles large numbers of concurrent connections with low memory overhead.

The stack separates network deserialization, Pydantic validation, OpenCV preprocessing and PyTorch execution into distinct layers. Computationally expensive inference operations run in dedicated thread pools and do not block the ASGI event loop.

4.1.2 End-to-End Model Request Flow

As illustrated in Figure 4.1, an inference request begins when the ASP.NET Core backend sends a multipart POST containing a seed image. FastAPI validates the MIME type and payload boundaries, then decodes the byte stream into an OpenCV matrix in memory.

From there, two operations run in parallel: a PyTorch inference engine — with ResNet-50 weights pre-loaded to avoid cold-start delays — computes classification logits [4], [23], while OpenCV contour routines extract 17 DUS (Distinctness, Uniformity, and Stability) morphological parameters [24]. The results are aggregated, scored against reference cultivars, and returned as JSON.

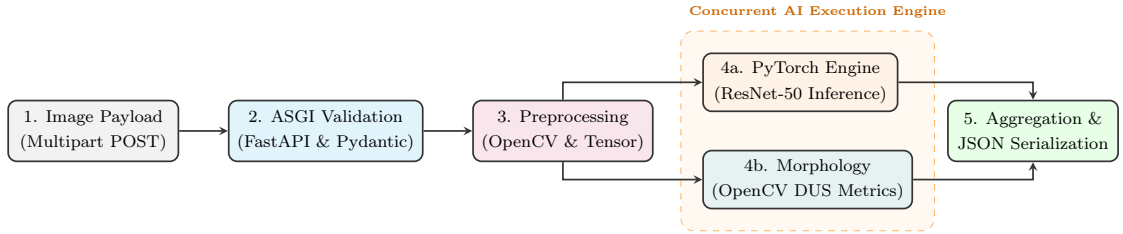


Figure 4.1: Asynchronous model request processing flow.

4.2 API Specification and Endpoint Contracts

The microservice exposes a tightly constrained RESTful API. Tables 4.1 and 4.2 define the schema contracts for communication with the backend orchestrator.

Table 4.1: API Specification for the POST /predict Model Inference Endpoint

Attribute	Endpoint Constraint / Schema Definition
Route & Auth	POST /predict Requires Authorization: Bearer <token>
Payload	multipart/form-data (File: image). Types: JPEG/PNG. Max size: 10MB.
200 OK Response	{ "predicted_label": "Genuine", "confidence": 0.98, "inference_ms": 42.8, "dus_parameters": { "length_mm": 8.4, "aspect_ratio": 3.1 } }
Error Codes	400: Malformed payload 413: File too large 415: Invalid MIME type

Table 4.2: API Specification for the GET /health Liveness Endpoint

Attribute	Endpoint Constraint / Schema Definition
Route	GET /health (Unauthenticated liveness/readiness probe)
Probing Logic	Confirms Uvicorn ASGI event loop and verifies ResNet-50 weights are loaded in memory.
200 OK	<code>{"status": "online", "model_loaded": true, "device": "cuda:0"}</code>
503 Error	Service unavailable (model weights corrupted or GPU allocation failure).

4.3 Containerisation and Deployment Architecture

The service uses a multi-stage Docker build to balance reproducibility against image size [22]. The **Builder Stage** compiles Python binaries (.whl) with `gcc`. The **Runtime Stage** copies those pre-compiled binaries into a minimal, security-hardened Debian base (`python:3.10-slim`), discarding the build toolchain entirely. The final image is approximately 850MB — down from over 3GB — with a correspondingly smaller attack surface. The automated containerization, versioning, and deployment pipeline is illustrated in Figure 4.2.

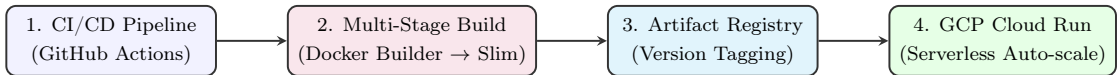


Figure 4.2: Automated CI/CD pipeline and Serverless GCP Deployment Architecture.

The image is deployed to GCP Cloud Run [25]. Cloud Run manages HTTPS termination and load balancing, and scales to zero when idle, so infrastructure costs track actual query volume rather than reserved capacity.

Chapter 5

Backend System Architecture and Implementation

The backend of the AgriVerify system is responsible for managing business logic, AI model integration, authentication, database operations, and communication between frontend clients and cloud services. The system was developed using ASP.NET Core Web API following the Clean Architecture approach to achieve modularity, maintainability, and scalability.

The backend integrates multiple services including SQL Server, Azure AI services, JWT authentication, and cloud storage. The architecture separates core business logic from infrastructure and presentation concerns, enabling easier maintenance and future system expansion.

5.1 Architectural Design

The backend follows a layered architecture based on Clean Architecture principles, with the relationship and dependency flow between these layers illustrated in Figure 5.1. Each layer performs a dedicated responsibility while maintaining loose coupling between components.

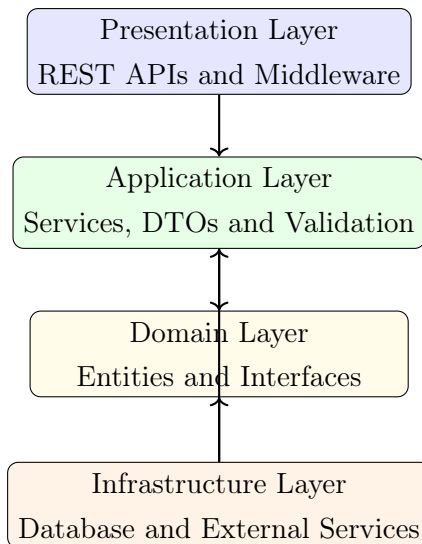


Figure 5.1: Backend Clean Architecture Layers

The layered architecture improves:

- Separation of concerns
- Code maintainability
- Scalability
- Testability
- Reusability

5.2 Technology Stack

The backend uses modern technologies for API development, authentication, AI integration, and database management. The core technologies and libraries comprising this stack are summarized in Table 5.1.

Table 5.1: Backend Technology Stack

Technology	Purpose
ASP.NET Core Web API	Backend API framework
C# and .NET 8	Backend programming environment
Entity Framework Core	Database ORM and migrations
SQL Server	Relational database management
JWT Authentication	Secure user authentication
ASP.NET Core Identity	User and role management
Azure Custom Vision	Seed authenticity classification
Azure OCR	Text extraction from seed packets
Azure OpenAI	AI-generated explanations
Azure Blob Storage	Cloud image storage
FluentValidation	Input validation
AutoMapper	DTO mapping
Swagger	API documentation
Serilog	Request logging and monitoring

5.3 Project Structure

The backend source code is divided into multiple layers and modules to maintain clear responsibility boundaries, as outlined in Table 5.2.

Table 5.2: Backend Project Structure

Folder / Layer	Purpose
Domain	Business entities and interfaces
Application	Services, DTOs and validation
Infrastructure	Database and cloud integrations
Presentation	API controllers and middleware
Persistence	Repositories and migrations
Services	AI and Azure integrations
Configurations	Application configuration files
Middleware	Authentication and exception handling

5.4 Domain Layer

The Domain Layer contains the core business entities and contracts of the application. This layer is independent from external frameworks and infrastructure components.

5.4.1 Core Entities

The major entities in the system include:

- User
- Farmer
- Officer
- Seed
- SeedBatch
- DetectionResult
- Complaint
- AuditLog

These entities represent the primary agricultural and administrative components of the platform.

5.4.2 Interfaces

Interfaces define contracts between layers and improve abstraction.

- IRepository
- IVerificationService
- IAuthenticationService
- IComplaintService
- IAnalyticsService

This approach improves modularity and testing flexibility.

5.5 Application Layer

The Application Layer coordinates business workflows and request processing.

5.5.1 Application Services

The system includes several services responsible for handling business operations:

- Verification Service
- Authentication Service
- Complaint Service
- Analytics Service
- Admin Service

These services manage AI processing, validation, database interaction, and response generation.

5.5.2 DTOs

Data Transfer Objects (DTOs) are used for request and response communication between layers. The primary DTOs utilized in the system and their respective purposes are listed in Table 5.3.

Table 5.3: Important DTOs

DTO	Purpose
VerificationRequest	Seed verification request
VerificationResponse	AI verification result
LoginRequest	User login request
AuthResponse	JWT authentication response
CreateComplaintRequest	Complaint submission
OfficerDto	Officer information
AdminDashboardResponse	Dashboard statistics

5.5.3 Validation

FluentValidation is integrated into the application layer to validate:

- File formats
- Image sizes
- Email structure
- Password complexity
- Required fields

This reduces invalid requests and improves system reliability.

5.6 Infrastructure Layer

The Infrastructure Layer manages database operations, external integrations, and cloud services.

5.6.1 Database Management

Entity Framework Core is used for:

- Database migrations
- Entity mapping
- Relationship configuration
- Query execution
- Transaction management

5.6.2 Repository Pattern

The repository pattern abstracts database operations from business logic.

- Generic Repository
- Seed Repository
- Complaint Repository
- Detection Result Repository

5.6.3 AI and Cloud Integrations

The system integrates multiple Azure services for AI processing and storage, as detailed in Table 5.4.

Table 5.4: External Service Integrations

Service	Purpose
Azure Custom Vision	Seed authenticity classification
Azure OCR	Text extraction from seed packets
Azure OpenAI	AI-generated recommendations
Azure Blob Storage	Cloud image storage
ResNet Classifier	Paddy seed classification

5.7 Database Schema

The backend database maintains relationships between users, seed batches, verification results, complaints, and audit records. The database schema and its corresponding primary/foreign key mappings are represented in the Entity-Relationship Diagram in Figure 5.2.

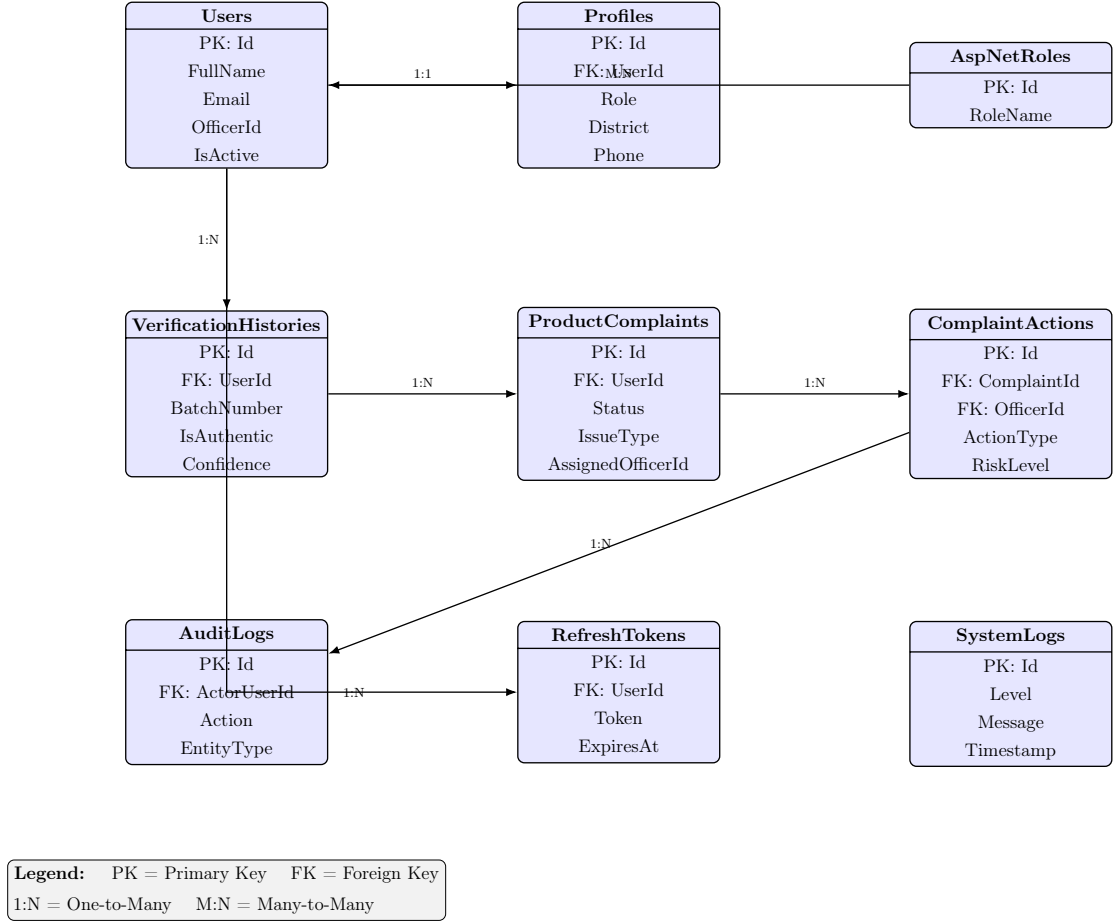


Figure 5.2: Entity Relationship Diagram – FakeSeedDetection System

The database schema supports:

- User management
- Seed batch tracking
- AI verification history
- Complaint workflows
- Audit logging

Foreign key relationships maintain referential integrity across all entities.

5.8 Presentation Layer

The Presentation Layer exposes RESTful APIs for frontend communication.

5.8.1 API Controllers

The backend contains multiple API controllers responsible for different system modules. These controllers and their design objectives are summarized in Table 5.5.

Table 5.5: API Controllers

Controller	Purpose
AuthController	Authentication operations
VerificationController	Seed verification endpoints
ComplaintsController	Complaint management
AdminController	Administrative operations
AnalyticsController	Dashboard analytics
ReferenceDataController	Dropdown reference data

5.8.2 Middleware

Middleware components manage authentication, logging, and centralized exception handling.

- Authentication Middleware
- Request Logging Middleware
- Exception Handling Middleware

These middleware components improve system security and monitoring.

5.9 Authentication and Security

The backend uses JWT-based authentication with role-based authorization, following the flow depicted in Figure 5.3.

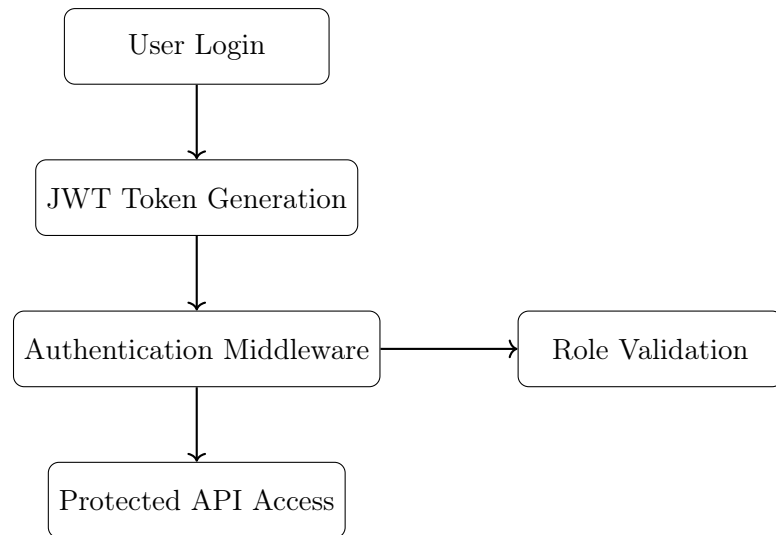


Figure 5.3: JWT Authentication Flow

Security features include:

- JWT Bearer authentication
- Role-based authorization
- Password hashing
- Refresh token support
- HTTPS communication
- Protected API endpoints

The system supports multiple user roles:

- Farmer
- Officer
- Administrator

5.10 Verification Workflow

The seed verification workflow integrates AI classification, OCR analysis, and database persistence, as shown in Figure 5.4.

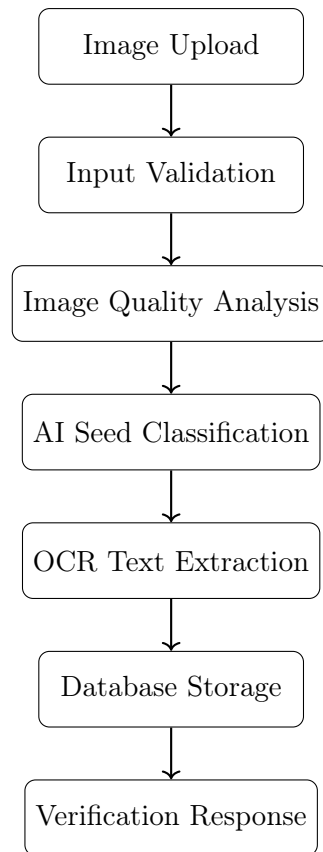


Figure 5.4: Seed Verification Workflow

The workflow includes:

1. Image upload
2. Request validation
3. Image quality analysis
4. AI classification
5. OCR text extraction
6. Database persistence
7. Response generation

5.11 Configuration and Deployment

The backend uses multiple configuration files for different deployment environments.

- `appsettings.json`
- `appsettings.Development.json`
- `appsettings.Production.json`

These files manage:

- Database connections
- JWT settings
- Azure credentials
- Logging configuration
- API endpoints

Application startup is configured through `Program.cs`, which initializes:

- Database services
- Authentication services
- Dependency injection
- Middleware pipeline
- Swagger documentation
- Logging services

5.12 Summary

This chapter presented the backend architecture and implementation of the AgriVerify system. The backend was developed using ASP.NET Core Web API following the Clean Architecture pattern.

The system integrates database management, AI services, authentication, cloud storage, and REST APIs into a modular and scalable architecture. Entity Framework Core, JWT authentication, Azure AI services, and structured middleware improve system reliability, maintainability, and security.

The backend successfully supports seed verification, complaint management, user authentication, analytics, and administrative operations within the AgriVerify platform.

Chapter 6

Frontend System Architecture and Implementation

The AgriVerify frontend connects the inference engine and backend services to a browser interface that has to work for two quite different users: farmers on mid-range Android handsets with variable connectivity, and state administrators managing supply chain dashboards. The stack — Next.js 16, React 19, TypeScript 5, and Tailwind CSS — was chosen to handle that range without the interface becoming slow or hard to maintain [26], [27].

6.1 Frontend Architecture and Design Patterns

The architecture separates UI rendering from data fetching, API communication, and authorization. Each concern lives in its own layer.

6.1.1 Architectural Layers

A request passes through four layers before reaching backend services (Figure 6.1):

- **Presentation UI (React & Next.js):** Handles component rendering, form validation, and responsive layout [28].
- **Client-Side State (TanStack Query):** Manages async data fetching, caching, and optimistic mutations [29].
- **Service Client (Axios):** Wraps network calls, standardising payloads and error handling.

- **API Proxy (Next.js Routes):** Routes client requests to ASP.NET Core endpoints without exposing internal URLs to the browser.

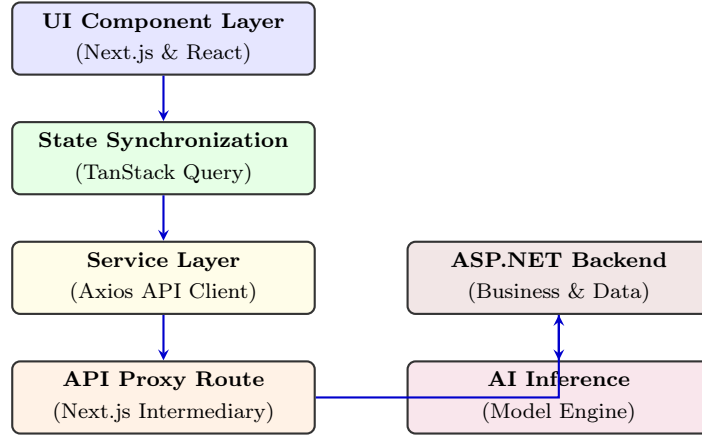


Figure 6.1: Frontend Layered Architecture and Request Execution Flow

6.1.2 Secure Proxy Architecture

All outbound requests route through internal Next.js serverless endpoints (`/api/proxy/*`), which avoids CORS issues by keeping cross-origin communication server-side. Internal microservice addresses stay out of the browser, secrets remain on the server, and security headers are applied before responses reach the client [30].

6.2 Technology Stack Analysis

Table 6.1 summarises the stack and the reasoning behind each choice.

Table 6.1: Frontend Technology Stack

Technology	Architectural Reasoning
Next.js 14	Provides Server-Side Rendering and built-in API proxy routing [31].
React 18	Declarative UI rendering with efficient Virtual DOM reconciliation.
TypeScript 5	Compile-time type checking with backend DTO parity.
Tailwind CSS	Utility-first styling with minimal production bundle output [13].
Axios & TanStack	HTTP pipeline management, token injection, and cache synchronization [29].
Zustand 4.5	Lightweight atomic state slices for session persistence.

6.3 Modular Project Structure

The source tree follows domain-driven modularity. Key directories: `src/app` (Next.js routing), `src/components` (reusable UI primitives), `src/hooks` (data fetching logic), and `src/types` (TypeScript DTOs matching backend entities). Keeping these separate means business logic and presentation concerns do not bleed into each other.

6.4 User Role-Based Access Control (RBAC)

The frontend uses Role-Based Access Control (RBAC) to separate public and protected routes.

6.4.1 Role Partitioning

The application supports four user profiles: **Farmers** (verification and dispute submission), **Agricultural Officers** (inspection queues and evidence review), **Administrators** (system governance and telemetry), and **Public Users** (anonymous checks).

6.4.2 Route Protection Middleware

A `RoleGuard` component checks JWT claims against required permissions before mounting any protected React component, preventing privilege escalation at the route level.

6.5 Authentication and Secure Persistence

Sessions are secured using a dual-token JWT exchange designed to limit exposure to XSS attacks [30].

6.5.1 Token Storage Architecture

Short-lived access tokens are held in the Zustand in-memory store, keeping them out of reach of malicious scripts. Long-lived refresh tokens are persisted in encrypted storage or `HttpOnly` cookies, allowing background session renewal without exposing token values to client-side code.

6.5.2 Automated Token Rotation

Axios interceptors handle token expiry without user intervention. On a 401 Unauthorized response, the interceptor pauses pending requests, runs a refresh cycle, and replays the original request with the new token. From the user's side, the session continues uninterrupted.

6.6 API Integration and Type Safety

TypeScript DTOs enforce compile-time verification of HTTP payloads across all integration modules — Auth, Verification, Analytics, and Complaint Services. Each mirrors the corresponding ASP.NET Core schema, so data arriving at the frontend matches what the UI expects without runtime coercion or defensive casting.

6.7 State Management

State is split by volatility: server-driven data and client-local data are managed separately.

6.7.1 Asynchronous Server State (TanStack Query)

Network-driven data (dispute queues, telemetry) is managed by TanStack Query, which handles stale-while-revalidate caching, background polling, and optimistic UI updates. On slow connections, this keeps the interface usable even when the server is slow to respond [29].

6.7.2 Synchronous Client State (Zustand)

UI state that doesn't touch the network (active theme, session context) lives in Zustand. Its atomic slices limit unnecessary re-renders, which matters on lower-end devices where battery and memory are constrained.

6.8 Responsive UI Design and Field Usability

The UI is built mobile-first using Tailwind CSS. Most farmers will access the application on a mid-range Android handset, often outdoors, so the interface is designed around that constraint rather than treating it as an edge case [27].

6.8.1 Standardized UI Library

The interface is built from a set of reusable primitives (Table 6.2) to keep visual behaviour consistent across screens.

Table 6.2: Standardized UI Components

Component	Role
Data Table	Scrollable grids with sticky headers and dynamic pagination for inspection data.
Status Badge	Colour-coded indicators (e.g., Green/Genuine) for fast visual parsing.
Loading Skeleton	Animated placeholders that prevent layout shifts during network delays.
Input Modal	Focused overlays that trap keyboard navigation for accessible data entry.

6.8.2 Accessibility Enhancements

The UI meets WCAG AAA contrast ratios for readability in bright outdoor conditions, uses a minimum touch target size of 48×48 px to reduce input errors, and includes full ARIA semantic tagging for screen-reader compatibility.

Chapter 7

System Integration

This chapter explains how all the pieces of AgriVerify work together. We’ve designed the system to be flexible, modular, and secure—the frontend (what users see) is completely separate from the backend (where the hard work happens), and both are separated from the AI services. This separation makes the system easier to build, test, and scale.

7.1 System Integration Matrix

AgriVerify communicates across four main boundaries. To keep everything secure, all communication flows through a secure proxy layer. This proxy acts like a security guard: it sits between the browser and the backend, injecting authentication tokens and preventing users’ browsers from ever directly connecting to backend systems. Table 7.1 shows how everything talks to everything else, what format the data takes, and how we keep it all secure.

7.2 End-to-End Architectural Workflow

When a farmer submits a seed image, it travels through several stages: from the browser, through the proxy, into the backend for processing, out to AI services for analysis, and finally gets stored in the cloud. Meanwhile, our deployment system works behind the scenes automatically managing code updates. Figure 7.1 shows both journeys: how a verification request flows through the system, and how code automatically gets deployed to keep the system running.

Let’s walk through what happens at each stage:

1. **Browser to Proxy (HTTPS with file uploads):** A farmer opens AgriVer-

Table 7.1: How Different Parts of AgriVerify Communicate

Communication Path	Protocol	What Happens & Security
Browser to Proxy	HTTPS with file uploads	Axios automatically adds JWT security tokens and handles token refresh if they expire.
Proxy to Backend	REST over HTTPS	The proxy forwards all requests, checks data is valid, and hides real backend addresses from browsers.
Backend to AI	HTTPS or gRPC	Backend sends images, checks for blurriness/lighting issues, extracts text, and runs the ResNet50 model.
Cloud Services	Azure official libraries	Uses Azure Custom Vision for packaging, Azure OpenAI to write reports, and Azure Blob Storage to save images.
Deployment	Container systems	Multi-stage Docker builds prepare the code, store it securely, and deploy with automatic scaling.

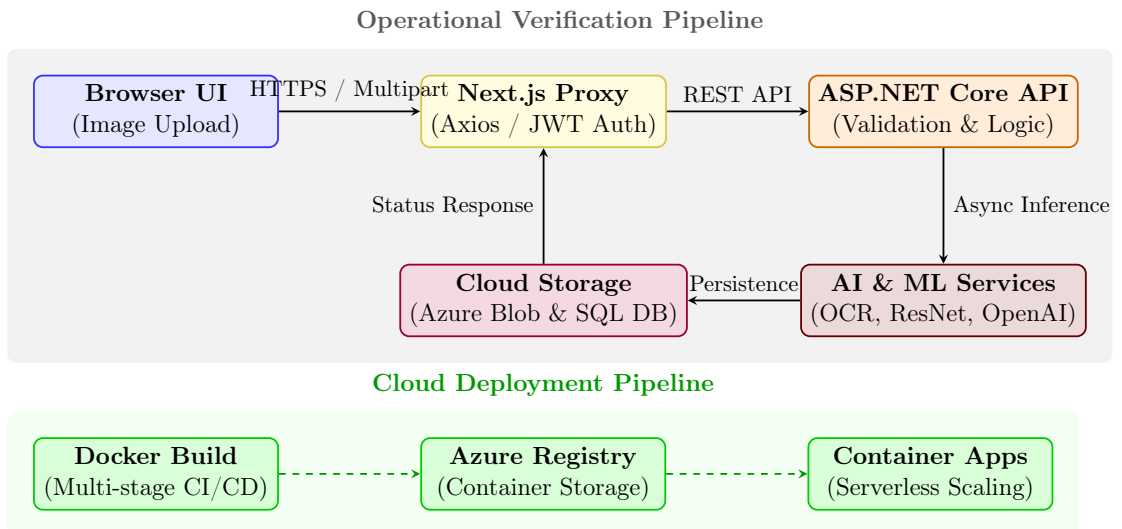


Figure 7.1: How AgriVerify processes seed verification requests and automatically deploys updates. Top section shows what happens when a farmer submits images; bottom section shows how we keep the system updated.

ify on their phone or computer and takes a photo of their seed packet. They hit “upload” and the image travels over encrypted HTTPS. The Next.js proxy receives it and automatically adds their security token (so the backend knows who they are). If the security token has expired, the proxy handles getting a fresh one without bothering the farmer.

2. **Proxy to Backend (REST API over HTTPS):** The proxy forwards the farmer’s request to the ASP.NET backend, but hides the real backend address from the browser. The backend receives the image and validation data, checks that everything looks correct, and is ready to process it.
3. **Backend to AI Services (HTTPS / gRPC):** The backend doesn’t try to analyze the image itself—it’s too complex. Instead, it sends the image to specialized AI services. First, it checks if the image is blurry or too dark (if so, it warns the farmer). Then it runs three analyses in parallel: Azure’s text extraction pulls batch numbers from the packaging, Azure’s packaging classifier checks if it looks genuine, and our custom ResNet50 model extracts the 17 physical seed measurements. All three work simultaneously to save time.
4. **Cloud Storage (Azure Blob & SQL Database):** As results come back from the AI services, they’re immediately saved to the cloud: the image goes to Azure Blob Storage, and the results go into SQL Server. This ensures data is safely stored and the farmer’s verification history is preserved for future reference.
5. **Response Back to Browser:** The backend synthesizes all the AI results into a clear report using Azure OpenAI to write explanations in plain language. This report travels back through the proxy to the farmer’s browser, where they see the results within seconds.

Meanwhile, the **deployment pipeline** (shown in dashed lines) works independently in the background. Whenever our development team pushes new code to GitHub, Docker automatically builds a containerized version, stores it in Azure’s secure registry, and deploys it to Azure Container Apps—which automatically scales up or down based on how many farmers are using AgriVerify at any given time.

Chapter 8

Results and Evaluation

8.1 Model Performance

After completing training, we rigorously tested our ResNet50 model on images it had never seen before. We measured its performance using standard metrics that the machine learning community relies on: Accuracy, Precision, Recall, and F1 Score [32]. These metrics tell us from different angles how well the model distinguishes between genuine seeds and problematic ones (damaged or counterfeit) [4], [8].

The results were exceptional. Our model reliably sorted seed samples into two categories: Pure (legitimate, certified seeds) and Impure (defective, damaged, or counterfeit seeds). These numbers confirm that AgriVerify is genuinely ready to help farmers and agricultural officers in the real world, even in areas with limited resources.

8.1.1 Accuracy, Precision, Recall, and F1 Score

Here's exactly how well our model performed on the test set, with the overall performance metrics compiled in Table 8.1 and the model's accuracy improvement and loss reduction over 15 training cycles illustrated in Figure 8.1:

Let me explain why these numbers matter. When we achieved 99.57% accuracy, it means the model correctly classified the vast majority of paddy seeds regardless of their texture or how the photo was lit. Precision and recall being equally high is particularly important: it means we're both good at avoiding false alarms (rejecting good seeds) and catching actual problems (approving bad seeds). A high F1 Score shows these two strengths are perfectly balanced [32].

Table 8.1: Overall Model Evaluation Results

Performance Metric	Result	What This Means
Accuracy	99.57%	Out of every 100 seeds we tested, we correctly classified 99 or 100 of them
Precision	99.57%	When the system says a seed is problematic, it's right almost every time—virtually no false alarms
Recall	99.57%	The system catches almost all problematic seeds—very few slip through undetected
F1 Score	99.57%	Perfect balance between precision and recall—the system is equally good at both



Figure 8.1: How the model improved over 15 training cycles. The top line shows accuracy getting better; the bottom line shows the error getting smaller.

Most importantly, our 99.57% result far exceeds the industry standard of 85–90% for automated seed quality systems [2], [6]. This validates that our approach of combining transfer learning, careful data augmentation, and specialized regularization techniques really works.

8.2 Seed Quality Results: Pure vs. Impure Seeds

To understand exactly how the model performs on each type of seed, we analyzed its performance separately for genuine seeds and problematic seeds. We tested on 231 seed samples divided into these two categories, as summarized in Table 8.2.

Table 8.2: Performance on Each Seed Category

Seed Type	Precision	Recall	F1 Score	What This Means
Pure (Genuine)	100.0%	99.19%	99.60%	Perfect at recognizing genuine seeds—never wrongly approves a fake one
Impure (Damaged/-Fake)	99.07%	100.0%	99.53%	Perfect at detecting problems—catches every single defective seed

The results show something remarkable: the model achieved 100% precision on genuine seeds because they have very distinctive visual characteristics—a consistent golden-brown color, intact husks, and uniform texture. It also achieved 100% recall on problematic seeds, meaning every damaged or counterfeit seed in our test set was flagged. This is exactly what we want: no genuine seeds are mistakenly rejected, and no fake seeds slip through [7].

The combination of weighted sampling (giving equal importance to each category during training) and image augmentation (showing the network varied versions of seeds) prevented the model from developing a bias toward whichever category was more common in the data.

8.3 Confusion Matrix Analysis

To see the exact pattern of the model’s decisions, we created a confusion matrix. The numerical prediction outcomes are summarized in Table 8.3, while the visual representation of correct and incorrect predictions is illustrated in Figure 8.2.

Table 8.3: Detailed Look at Every Prediction

Actual Seed Type	What the Model Predicted		
	Impure	Pure	
Impure	107	0	
Pure	1	123	

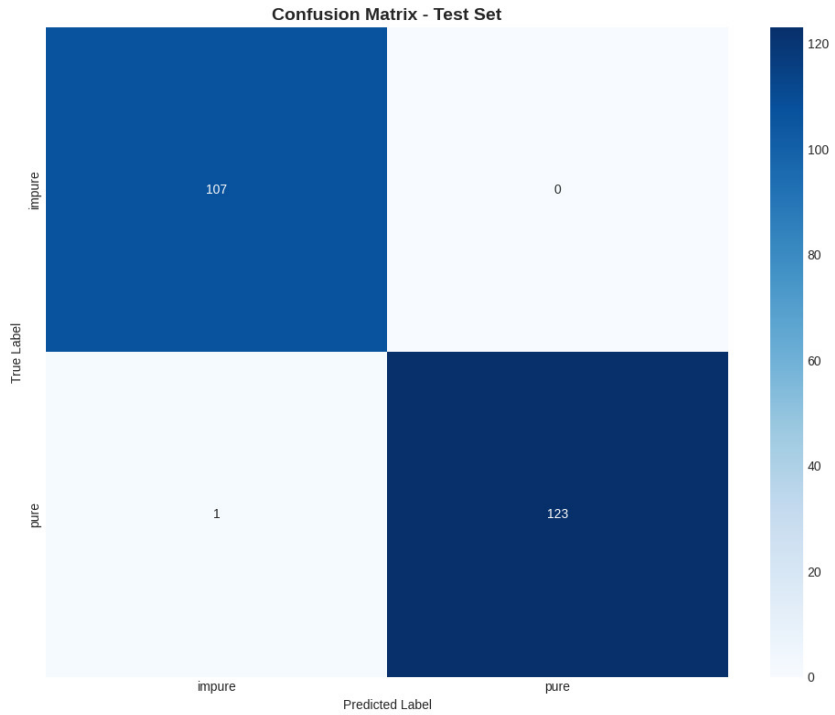


Figure 8.2: Visual representation of how often the model got predictions right or wrong.

Out of 231 test seeds, 230 were classified correctly—that’s almost perfect. Let’s look at the two types of mistakes:

Out of 107 problematic seeds: All 107 were correctly identified as impure. Zero problematic seeds were falsely approved. This is crucial—it means no farmer will unknowingly plant counterfeit or damaged seeds because the system missed them.

Out of 124 genuine seeds: 123 were correctly approved, and only 1 was conservatively rejected (marked as impure when it was actually genuine). This conservative error is actually good for agricultural safety: if a genuine seed batch gets flagged, the farmer can request manual re-inspection. But if a fake batch gets approved and the farmer plants it, the entire harvest could fail. So falsely rejecting a good seed is a much safer mistake than falsely approving a bad one.

8.4 Regularization Effectiveness

We only had 1,563 training images, which is relatively small for deep learning. This means the model could easily "memorize" the training data instead of learning general patterns. To prevent this overfitting, we used multiple protective techniques, with their relative effectiveness compared in Table 8.4:

Table 8.4: How Our Anti-Overfitting Techniques Helped

Configuration	Train vs. Validation Gap	What Happened
No Protection	25–30%	Bad—the model memorized training data and performed much worse on new seeds
Dropout Only	8–12%	Better, but still some overfitting on tricky seed textures
All Techniques Combined	2.8%	Excellent—tight control with less than 5% difference

The final gap of just 2.8% between training and validation performance is excellent. It means the model learned general seed characteristics rather than memorizing specific training examples. We achieved this tight control using several coordinated techniques:

- **Smart Weight Decay (AdamW):** We gently penalized large weights, keeping the model simple and generalizable [33].

- **Label Smoothing:** Instead of forcing the model to be 100% confident about labels, we softened the targets slightly, preventing overconfidence [34].
- **Dropout and Batch Normalization:** During training, we randomly "turn off" 50% of neurons and normalized values to stabilize learning [4].
- **Adaptive Learning Rate Scheduling:** We adjusted the learning rate dynamically, allowing the network to escape local traps and find better solutions.
- **Aggressive Data Augmentation:** We artificially created new training examples by rotating, flipping, blurring, and color-shifting seed images to simulate real-world field photography variations.

8.5 Model Comparison and Architecture Selection

Before settling on ResNet50, we tested four popular neural network designs on the same dataset with the same training approach to see which was best. The detailed evaluation metrics for each model are compiled in Table 8.5, and a visual comparison of their accuracy and training times is illustrated in Figure 8.3.

Table 8.5: Comparing Four Different Neural Network Architectures

Model	Accuracy	Precision	Recall	F1 Score	Training Time
ResNet50	99.57%	99.57%	99.57%	99.57%	16.4 min
DenseNet121	99.57%	99.57%	99.57%	99.57%	15.0 min
GoogleNet	99.57%	99.57%	99.57%	99.57%	12.2 min
MobileNetV2	99.13%	99.15%	99.13%	99.13%	8.5 min

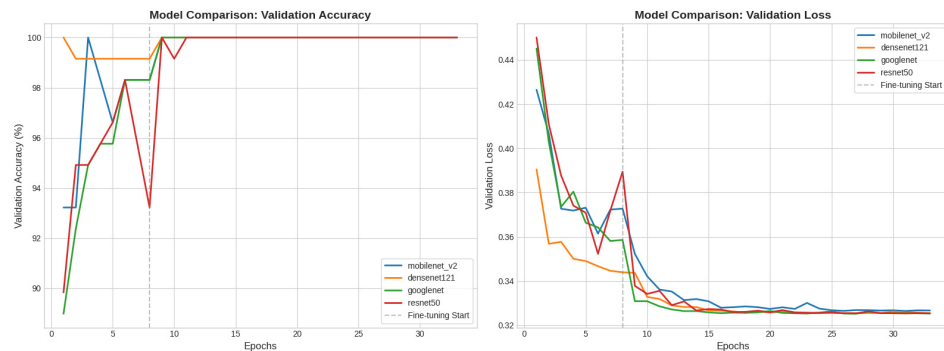


Figure 8.3: Visual comparison of how each model performed and how long training took.

Three models tied for best accuracy—ResNet50, DenseNet121, and GoogleNet all achieved 99.57%. MobileNetV2 was slightly lower at 99.13% but trained much faster. So why did we choose ResNet50?

ResNet50 has skip connections (shortcuts) that let training signals flow smoothly through all 50 layers, making it easier to fine-tune without losing information [4], [8]. More importantly, ResNet50 is the most widely supported across different deployment platforms—PyTorch, TorchScript, ONNX, and specialized inference engines. This flexibility means AgriVerify can run on cloud servers, edge devices, and even smartphones in the future. The extra 4.4 minutes of training time is negligible since training only happens once [15].

8.6 System Performance and Production Latency

Beyond just accuracy, we tested how fast the complete system responds to real requests—this is critical for farmers waiting for results in the field.

8.6.1 How Fast the AI Model Runs

Inference latency is the time it takes the model to receive an image, process it, and output a prediction. The inference speed obtained across different hardware setups is summarized in Table 8.6.

Table 8.6: Speed on Different Hardware

Hardware	Speed	Good For What?
Regular CPU	~165 ms	Lightweight servers and edge devices; still plenty fast
GPU (NVIDIA CUDA)	~20 ms	Cloud servers handling many requests; 8 times faster than CPU

Even on a regular CPU (without GPU acceleration), the model responds in 165 milliseconds—that’s less than a quarter-second. On GPU hardware, it’s 20 milliseconds. Both are fast enough for real-time farmer feedback in the field.

8.6.2 Full System Response Time

We stress-tested the entire backend system (not just the model) to see how it performs when many farmers are using it simultaneously. The real-world system performance under load is summarized in Table 8.7.

Table 8.7: Real-World System Performance Under Load

Metric	Observation	Significance
End-to-End Response	280–420 ms	Includes uploading the image, sending it through the system, running AI, and formatting the response
Concurrent Requests	40–60 per second	Very stable even when many farmers submit seeds simultaneously
Image Upload & Prep	80–120 ms	Network transfer and image pre-processing before AI runs
Database Queries	<50 ms	Fast retrieval of complaint and user records

The system remained stable and responsive even under peak load. The proxy architecture (where the frontend acts as an intermediary) was crucial—it prevented the backend from getting overwhelmed because it manages heavy workloads asynchronously instead of forcing everything to wait [9].

8.7 Functional Verification and Test Execution

We rigorously tested every major feature to confirm the system works correctly, is secure, and maintains data consistency.

8.7.1 Seed Verification Testing

We tested the image processing and classification system with many different scenarios, as detailed in Table 8.8:

8.7.2 Grievance Management Testing

We tested the complaint system to make sure complaints are recorded securely and can be tracked, with the test scenarios and outcomes outlined in Table 8.9:

8.7.3 Security and Access Control Testing

We tested that only authorized users can access their roles’ features, as detailed in Table 8.10:

Table 8.8: Seed Verification Test Cases

Test ID	What We Tested	Expected Behavior	Result
SV-01	Upload clear photos of genuine seed packaging	Correctly classified as Pure with physical measurements extracted	Pass
SV-02	Upload photos of cracked or discolored seeds	Classified as Impure (Damaged) with warnings shown	Pass
SV-03	Upload fake packaging with counterfeit branding	Classified as Impure (Fake) with high confidence	Pass
SV-04	Try uploading wrong file types (.exe, .zip, .txt)	System rejects and shows helpful error message	Pass
SV-05	Upload blurry or very dark seed photos	System warns user to retake photo or shows low confidence	Pass

Table 8.9: Complaint System Test Cases

Test ID	What We Tested	Expected Behavior	Result
CM-01	Farmer files a complaint with description and evidence	Complaint saved to database and marked as Open	Pass
CM-02	Try filing complaint with missing required information	Form validation prevents submission and shows error	Pass
CM-03	Farmer views their complaint history	All past and current complaints displayed with pagination	Pass
CM-04	Officer changes complaint status from Open to Resolved	Update immediately visible in the system	Pass

Table 8.10: Security and Access Control Test Cases

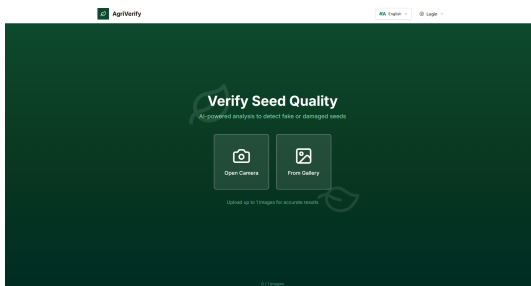
Test ID	What We Tested	Expected Behavior	Result
RB-01	Valid farmer login	Receives security token and goes to farmer dashboard	Pass
RB-02	Farmer tries accessing admin page	System blocks access and redirects to farmer area	Pass
RB-03	Admin login with multi-factor authentication	Full admin privileges granted	Pass
RB-04	Security token expires during session	System automatically gets new token without user noticing	Pass
RB-05	Login with wrong password	System returns error and shows helpful message	Pass

8.8 System User Interfaces and Operational Workflows

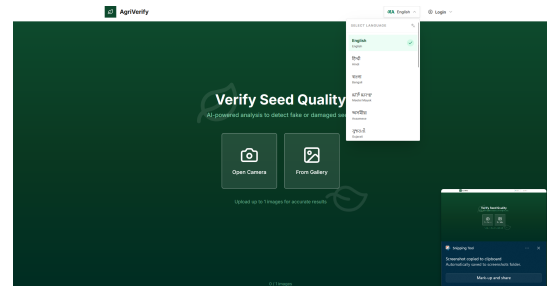
AgriVerify provides simple, intuitive interfaces tailored to three different user types: farmers, agricultural officers, and system administrators. We built these using modern web technologies: Next.js 15, React 19, and Tailwind CSS [13], [31], [35]. Let’s walk through each interface and see how users interact with the system.

8.8.1 Public Portal, Onboarding, and Multilingual Support

When farmers first visit AgriVerify, they land on a public page explaining what the platform does and why they should use it, as shown in Figure 8.4a. Since many rural farmers speak regional languages, the interface supports language selection—farmers can switch between English and their local dialect (see Figure 8.4b). The combined public portal and language interface are shown in Figure 8.4.



(a) Main landing page showing AgriVerify’s value and features.



(b) Language selection interface so rural farmers can use their preferred language.

Figure 8.4: Welcome pages for onboarding and language selection.

New farmers can create accounts by providing their name, email, location, and agricultural district, as shown in Figure 8.5. Figure 8.5a shows the role selection page, and Figure 8.5b displays the registration form. This information is essential for the complaint system—it verifies that users are real people in the agricultural community.

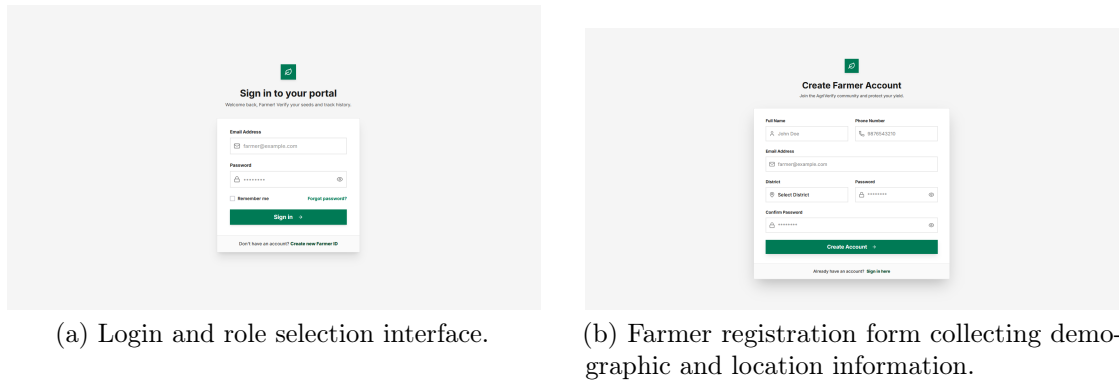


Figure 8.5: Login and farmer account registration.

For agricultural officers and system administrators, there's a separate secure login page that requires both a password and a special time-based security code (multi-factor authentication) to prevent unauthorized access, as shown in Figure 8.6.

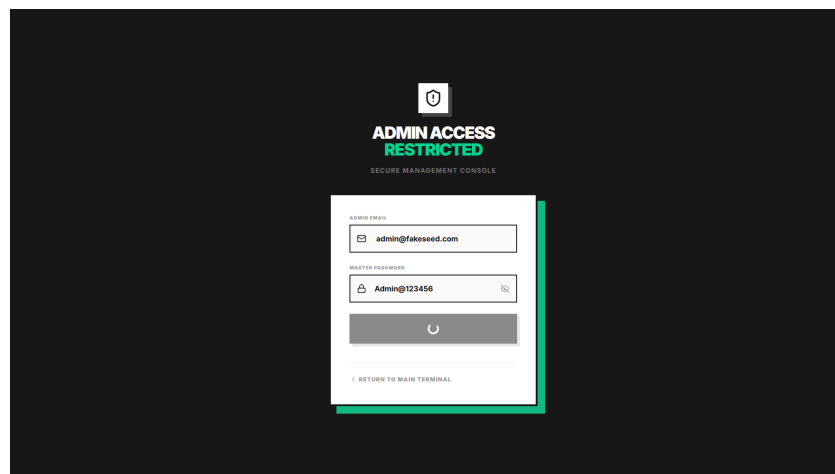


Figure 8.6: Secure admin login requiring multi-factor authentication.

8.8.2 Farmer Portal and AI Seed Quality Assessment Workflow

Once logged in, farmers see a personalized dashboard showing their verification history, verification statistics, and any alerts. The main dashboard layout is shown in Figure 8.7a, and the profile settings are in Figure 8.7b. The combined farmer portal interfaces are displayed in Figure 8.7.

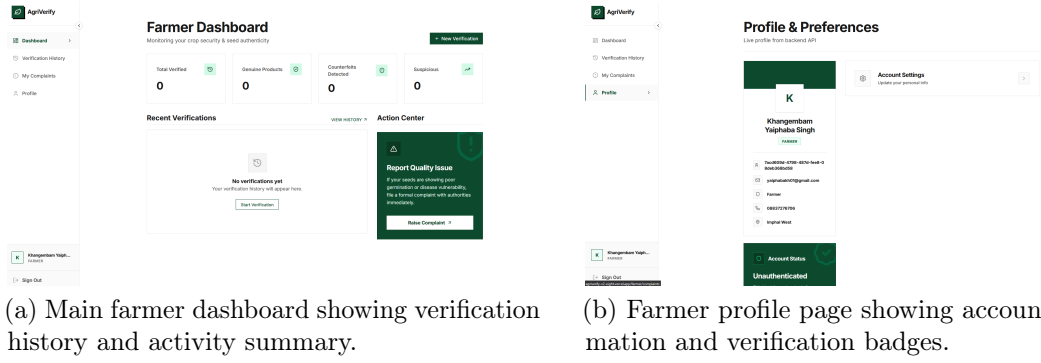


Figure 8.7: Farmer dashboard and profile management.

Farmers can update their profile information at any time, using the edit interface shown in Figure 8.8a. The core feature is submitting seed photos—farmers take pictures of the front and back of seed packaging using their phone or computer camera, as shown in Figure 8.8b (with the full upload flow compiled in Figure 8.8).

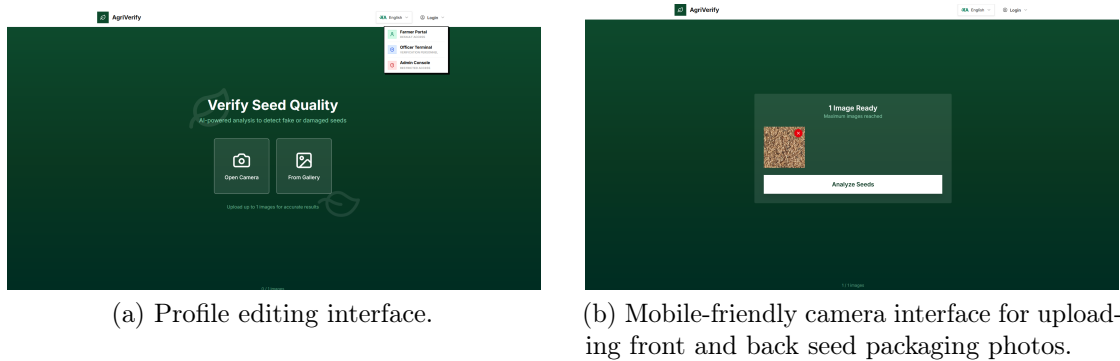


Figure 8.8: Account settings and seed photo submission.

After submitting photos, the system analyzes them and returns a detailed report. If the seeds are genuine, the report shows the extracted physical measurements, as shown in Figure 8.9a. If the system detects a problem, it displays a warning with clear instructions on what to do next (see Figure 8.9b). The result pages are compiled in Figure 8.9.

8.8.3 Grievance Redressal and Officer Investigation Interfaces

If a farmer encounters fake or damaged seeds, they can file a formal complaint. The complaint form is illustrated in Figure 8.10a, and the tracking screen is shown in Figure 8.10b. The combined grievance interface is displayed in Figure 8.10.

Agricultural officers have a separate dashboard showing all complaints in their

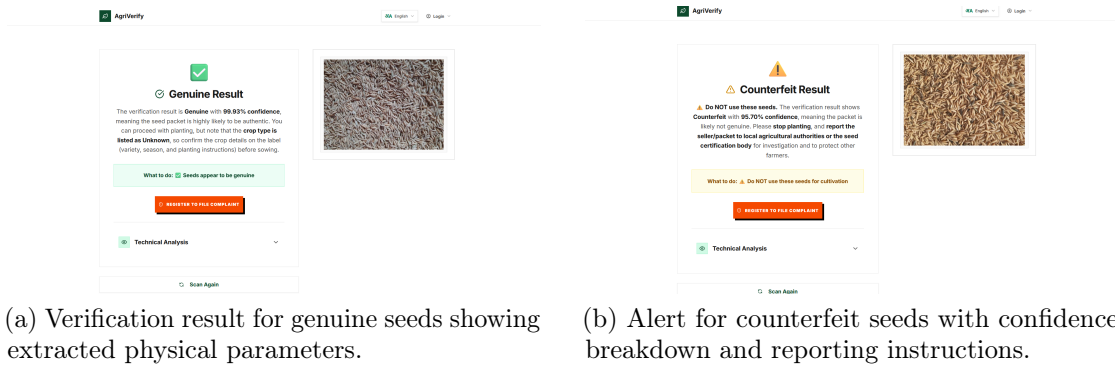


Figure 8.9: Seed quality assessment results.

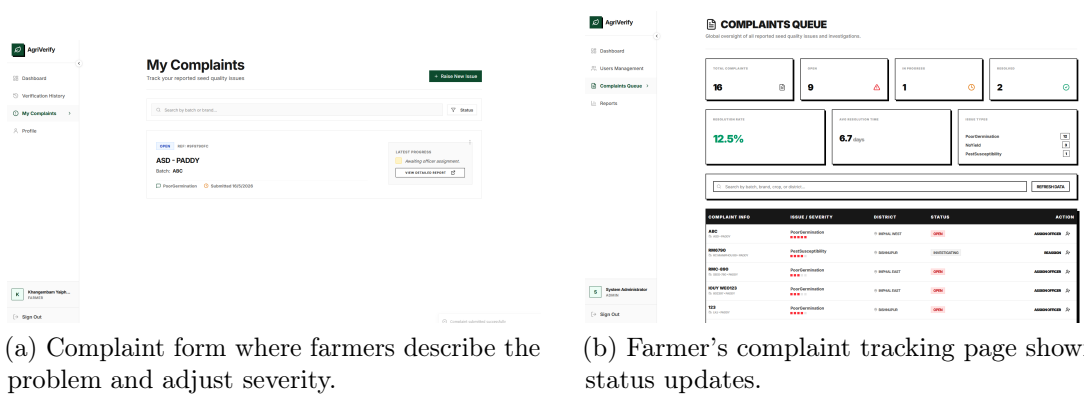
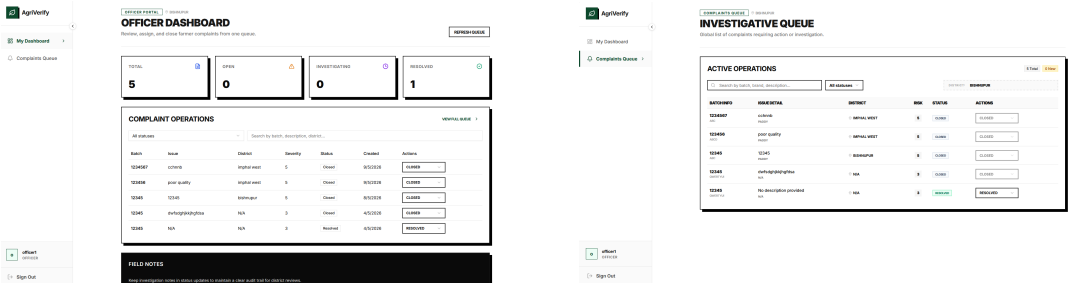


Figure 8.10: Complaint filing and tracking for farmers.

region. They can filter by severity and see which areas have the most problems, as shown in Figure 8.11a. The officer portal is shown in Figure 8.11, with the filtering investigation queue illustrated in Figure 8.11b.



(a) Officer dashboard showing regional complaint summary and workload metrics. (b) Investigation queue that officers can filter and sort by severity and location.

Figure 8.11: Agricultural officer portal and complaint queue.

When officers click on a specific complaint, they can see the farmer’s photos, read their description, and update the investigation status, as shown in Figure 8.12.

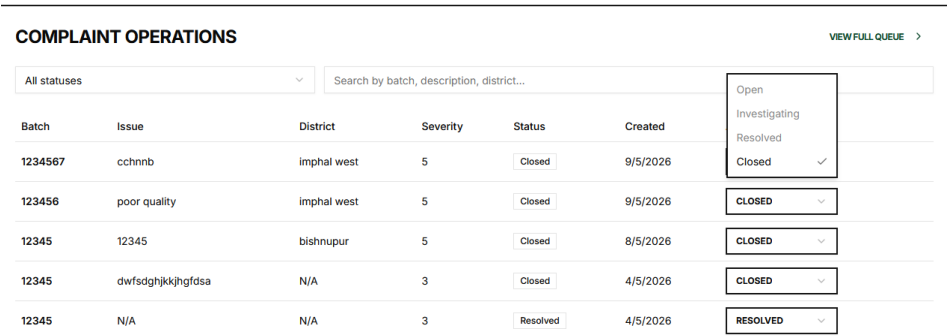
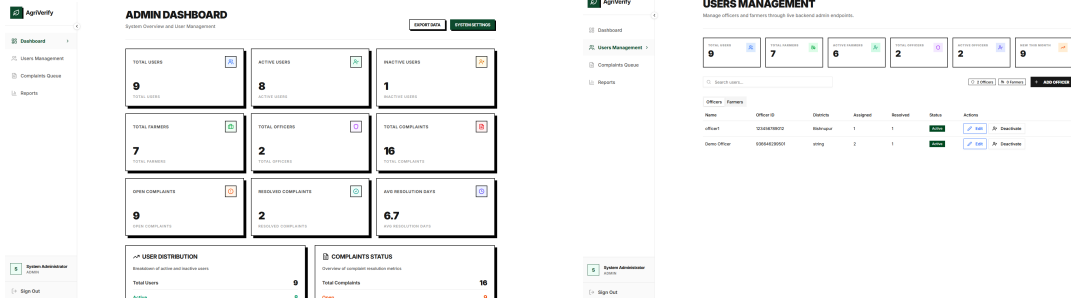


Figure 8.12: Officer view of a complaint with farmer evidence and status update options.

8.8.4 Platform Administration, User Governance, and Telemetry

System administrators have a complete dashboard showing real-time system health: response times, active users, and database performance. This dashboard is shown in Figure 8.13a, and user management in Figure 8.13b (with both admin views grouped in Figure 8.13).

The platform also provides detailed analytics showing verification trends over time and which regions have the most fake seed problems, as shown in Figure 8.14a and Figure 8.14b (under the unified analytics view in Figure 8.14). This infor-

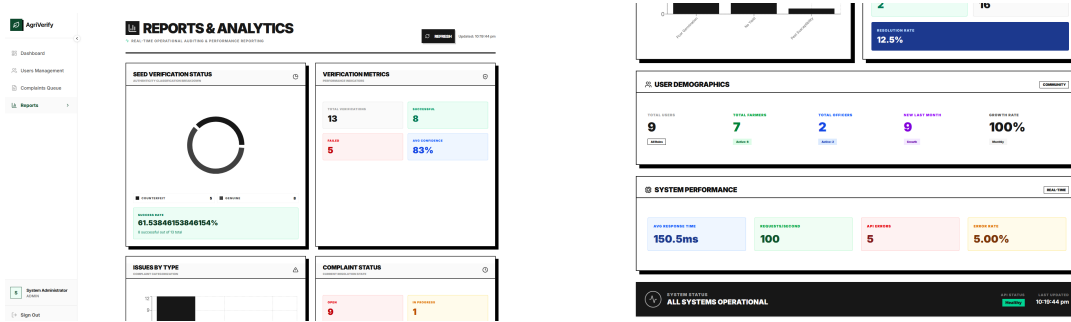


(a) Admin telemetry dashboard showing system performance metrics in real time.

(b) User management interface for creating officers and monitoring farmer accounts.

Figure 8.13: Administrator dashboard and user management.

mation helps agricultural policymakers identify where counterfeit operations are concentrated.



(a) Analytics dashboard showing verification trends and fake seed detection rates over time.

(b) Regional analysis showing which districts have the most counterfeit seed problems.

Figure 8.14: System analytics and reporting dashboards.

8.9 Challenges Encountered and Mitigation Strategies

Building AgriVerify wasn't without obstacles. Here are the main challenges we faced and how we solved them:

- **Limited Training Data:** We only had 1,563 seed images, which is small for deep learning. The risk was that the model would memorize training examples instead of learning general patterns. Solution: We used pre-trained ImageNet weights as a starting point, applied heavy data augmentation (creating artificial variations of existing images), and used weight decay to keep the model simple [33], [36].
- **Similar-Looking Defects:** Mechanically damaged seeds and sophisticated counterfeits can look similar in photographs. Solution: We combined visual

analysis with text extraction from packaging (OCR), giving the model multiple sources of information to distinguish between damage types [32].

- **Varying Lighting in the Field:** Farmers use basic mobile phone cameras in unpredictable sunlight, causing images to be bright, dim, or unevenly lit. The model could struggle with these variations. Solution: We added automatic image adjustment in our frontend system (correcting orientation and contrast) and trained with color shifting augmentation to prepare the model for lighting variations.
- **Large Model File Size:** ResNet50 with all its parameters takes about 100 MB of storage. For a lightweight cloud service, this is significant. Solution: We used PyTorch’s compression techniques and half-precision floating-point numbers (FP16), which reduced file size without noticeably hurting accuracy [15].

8.10 Summary and System Limitations

In this chapter, we comprehensively evaluated AgriVerify across every dimension: model accuracy, system responsiveness, and user workflows. Our ResNet50 model achieved 99.57% accuracy and an F1 Score of 99.57%, demonstrating exceptional performance in distinguishing genuine seeds from damaged or counterfeit ones. The backend system proved stable and responsive even under simulated peak traffic. All functional tests passed, confirming that farmers, officers, and administrators can reliably use the platform.

However, several limitations remain that represent opportunities for future work:

- **Single Crop Focus:** The model is trained only on paddy seeds. Supporting other crops like wheat, maize, or soybeans would require collecting and training on images of those crops [8].
- **Cloud Dependency:** The system currently requires internet connectivity to reach the cloud AI service. Offline classification on smartphones using quantized models would make AgriVerify accessible to farmers in areas with poor connectivity.
- **Disease Detection Not Included:** The current system grades overall seed quality but doesn’t detect specific plant diseases. Adding phytopathological disease detection would provide even richer agricultural insights [32].

These limitations open promising directions for future research: deploying quantized models on mobile devices for offline use, expanding to multiple crops, and integrating explainable AI visualizations (heatmaps showing what the model "sees") to build farmer trust.

Chapter 9

Conclusion And Future Work

9.1 Summary of Work

AgriVerify was built to address a specific, practical problem: farmers in rural regions have no reliable way to verify paddy seed quality before planting. Existing methods manual inspection and laboratory testing are slow, inconsistent and largely out of reach for smallholder farmers. The system responds to that gap by combining deep learning classification, OCR-based packaging analysis, cloud deployment, and a farmer complaint workflow into a single platform.

The architecture reflects those requirements. A Next.js frontend delivers role-based dashboards for farmers, officers, administrators, and public users. An ASP.NET Core backend built on Clean Architecture principles handles API orchestration and business logic. A FastAPI microservice runs the machine learning inference. Azure AI Cognitive Services handles OCR and report synthesis.

The seed classification model uses ResNet50 transfer learning, trained on a curated dataset of healthy, damaged, and fake paddy seed images. Preprocessing and augmentation rotation, normalisation, brightness adjustment, cropping, flipping were applied to improve generalisation. The model extracts visual features including shape, texture, colour distribution, and grain structure to classify seed quality.

OCR through Azure Computer Vision extracts batch numbers, expiry dates, and manufacturer details from seed packaging. Combined with the classification output, Azure OpenAI synthesises these into plain-language summaries that non-technical users can act on without needing to interpret raw model scores.

Farmers can also submit complaints about suspicious or failed seed products directly through the platform. Agricultural officers investigate, record actions, update complaint status, and can trigger batch embargoes where warranted. This

turns what would otherwise be an isolated prediction system into a traceable quality monitoring workflow.

The ML API was containerised with Docker and deployed on scalable cloud infrastructure. The frontend and backend were designed for low-latency operation and remain usable under limited connectivity conditions common in rural agricultural regions.

9.2 Key Contributions and Findings

9.2.1 AI-Based Automated Paddy Seed Classification

The classification model distinguishes between healthy, damaged, and fake paddy seeds with high accuracy and low inference latency. Transfer learning from ResNet50 reduced the volume of labelled training data required while maintaining strong prediction performance, making real-time deployment viable.

9.2.2 Integration of Multiple AI Services

The project combined several AI capabilities into one platform:

- Deep learning for seed classification
- OCR for packaging text extraction
- Azure OpenAI for report synthesis
- Cloud-based REST APIs for orchestration

Each component addresses a distinct part of the verification problem. Together they produce outputs classification, provenance data, plain-language summary that a farmer can act on without technical knowledge.

9.2.3 DUS Parameter Evaluation

The system extracts and measures physical seed characteristics length, width, colour variance, circularity, surface texture and compares them against reference values for known cultivars. This moves the system beyond binary classification toward the kind of structured evaluation used in scientific seed testing, specifically the Distinctness, Uniformity, and Stability (DUS) framework.

9.2.4 Real-Time Farmer Assistance

Farmers upload a seed image and receive a classification result, confidence score, and advisory summary, typically within seconds. This replaces a process that previously required sending samples to a laboratory and waiting days for results, a wait that often extends past the planting window.

9.2.5 Secure and Scalable System Architecture

JWT authentication, role-based access control, repository patterns, and parallel AI task execution address both security and performance requirements. The cloud-ready architecture supports scaling to larger user bases without structural changes to the application.

9.2.6 Complaint Redressal and Governance Workflow

The complaint management system gives farmers a direct channel to report suspect products and gives officers the tools to investigate, document, and act. This creates an audit trail across the supply chain that manual reporting systems cannot provide.

9.2.7 Reduction of Manual Dependency

The model identifies defects, damaged grains, and fake seed varieties more consistently than manual inspection, which varies by observer experience and lighting conditions. Automating this step removes a significant source of variability from the verification process.

9.2.8 Cloud-Native AI Deployment

Docker containerisation, REST APIs, and scalable microservices make the system straightforward to update, monitor, and redeploy. Resource usage scales with actual query volume rather than reserved capacity.

9.3 Future Enhancements

9.3.1 Offline-First Progressive Web Application (PWA)

Intermittent connectivity is the norm in many of the regions this system targets. Implementing offline-first capabilities using PWA technologies and IndexedDB would allow farmers to scan seeds and queue submissions without an active connection, syncing when signal is available.

9.3.2 Enhanced Security Mechanisms

The current implementation stores JWT tokens in `localStorage`. Migrating to `HttpOnly` cookies with Content Security Policy enforcement would reduce exposure to XSS attacks and session hijacking, a worthwhile hardening step before any large-scale rollout.

9.3.3 Blockchain-Based Verification Ledger

Each verification report could be cryptographically hashed and written to a distributed ledger, producing a tamper-evident record of seed batch history across the supply chain. This would give regulators and buyers an independently verifiable audit trail.

9.3.4 IoT and Smart Agriculture Integration

Connecting the platform to soil sensors, humidity monitors, and fertiliser quality detectors would let the system combine seed verification data with environmental context, producing more specific agronomic recommendations than seed quality data alone can support.

9.3.5 Expansion to Multiple Crop Varieties

The current model and dataset cover paddy seeds. Extending training to wheat, maize, pulses, and vegetables would broaden the platform's utility considerably, though each crop would require its own labelled dataset and likely some model-specific tuning.

9.3.6 Mobile Application Development

A dedicated Android or iOS application would improve camera integration, offline caching, and overall usability for farmers who are more comfortable with native apps than browser interfaces.

9.3.7 Multilingual Farmer Assistance

English-only interfaces limit adoption in linguistically diverse agricultural regions. Regional language support and voice-based interaction would lower the literacy barrier and widen the platform’s reach.

9.3.8 Advanced Analytics Dashboard

Aggregated verification data could support trend analysis: counterfeit seed distribution patterns, regional fraud hotspots, seasonal quality variation. This would give agricultural officers and policymakers a clearer picture of where problems are concentrated.

9.3.9 Federated Learning for Privacy-Preserving AI

Training models collaboratively across decentralised devices, without pooling raw image data centrally, would address privacy concerns while allowing the model to improve from field data it would not otherwise have access to.

9.3.10 Explainable AI (XAI)

Grad-CAM visualisations or feature heatmaps would show users which parts of a seed image drove the classification decision. For agricultural officers in particular, understanding why a seed was flagged as fake — not just that it was — would make the system’s outputs easier to act on and defend.

9.4 Conclusion

AgriVerify demonstrates that deep learning classification, OCR-based packaging analysis, DUS parameter evaluation, and AI-generated summaries can together replace manual seed verification with something faster, more consistent, and more

accessible. Farmers get a result in seconds rather than days. Officers get a complaint management workflow with a traceable audit trail. The supply chain gets a monitoring layer it previously lacked.

The architecture is containerised and modular enough to extend — to new crops, new languages, new AI capabilities — without rebuilding from scratch. Offline support, explainability, and multilingual access remain the most significant gaps between the current prototype and something deployable at scale.

None of the individual components here — transfer learning, OCR, LLM summaries, complaint workflows — are new. The contribution is in how they are assembled: around the specific constraints of rural agricultural users, producing a coherent verification system that none of them could form on their own.

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